INCH-POUND

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4 May 1992
SUPERSEDING
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23 August 1984

MILITARY SPECIFICATION

ERECTION AND MAINTENANCE EQUIPMENT FOR CABLE REINFORCEMENT SET FOR PANEL BRIDGE, M-2 (BAILEY TYPE)

This specification is approved for use within the USA Belvoir Research Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers all equipment required to install and maintain the cable reinforcement equipment for panel bridge M-2 (Bailey type) conforming to MIL-E-52714(ME).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-T-60	-	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-C-16173	-	Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-B-43436	-	Band, Marker, Blank.
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52714	-	Equipment, Cable Reinforcement, for Panel Bridge, M-2 (Bailey Type).
MIL-C-52950	-	Crate, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.
MIL-STD-1261	-	Arc Welding Procedures for Constructional Steels.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

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DRAWINGS

ME

TA13217E2171	- Erection Equipment for Set, Cable Reinforcement, for Panel Bridge M-2 (Bailey Type).
TA13217E2172	- Maintenance Equipment for Set, Cable Reinforcement, for Panel Bridge M-2 (Bailey Type).

(Copies of drawings, required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-JB, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.1	- Electrodes, Arc Welding, Covered Carbon Steel.
A 5.5	- Electrodes, Arc Welding, Low Alloy Steel Covered.
A 5.17	- Electrode and Fluxes, Carbon Steel, for Submerged Arc Welding.
A 5.18	- Welding, Carbon Steel Filler Metals for Gas Shielded Arc.
D 1.1	- Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association Inc., Attn: Traffic Order Section, 2200 Mill Rd., Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Applications for copies should be addressed to Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The erection equipment shall be in accordance with TA13217E2171 and as specified herein. Maintenance equipment shall be in accordance with TA13217E2172 and as specified herein. The contractor shall furnish all the erection equipment, all the maintenance equipment, or separate individual components selected from either of the two top assemblies as specified (see 6.2).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the end product. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2.1 Hydraulic-system components. In addition to the first article, all hydraulic components shown on the drawings shall be furnished and tested by the contractor prior to the time the first article is furnished. Tests shall

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be those specified in 4.5.2.1, 4.5.2.2, 4.5.2.3 and 4.5.2.5 as applicable and shall be subject to surveillance and approval of the Government.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and prevention control. The erection and maintenance equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the erection and maintenance equipment may be newly fabricated from recovered materials to the maximum extent practicable, provided the item produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the item.

3.3.3 Hydraulic oil. Hydraulic oil shall conform to MIL-L-46167.

3.3.4 Polytetrafluorethylene tape. Polytetrafluorethylene tape shall be commercial type and shall withstand temperatures in the range from -65 °F to +200 °F without deterioration. The tape shall be the type that allows threaded connections to be disconnected for maintenance.

3.3.5 Welding electrodes.

3.3.5.1 High strength steel. Welding electrodes for high strength steel shall be as specified on the drawings.

3.3.5.2 Low-alloy and structural steel. In addition to the welding electrodes specified on the drawings, welding electrodes for low-alloy and structural steel shall conform to either of the following AWS specifications:

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Specifications	Classification
A 5.1, A 5.5, A 5.17, A 5.18	E60XX, as applicable

3.4 Hydraulic system.

3.4.1 Component cleanliness. All hydraulic components, including hose assemblies, shall be cleaned prior to assembly by flushing.

3.4.2 Hydraulic cylinders. When tested in accordance with 4.5.2.3 and 4.5.2.4 the hydraulic cylinders shall show no evidence of external leakage (except weepage normally expected at piston rod packing), permanent deformation, or rupture of any part.

3.4.3 Hydraulic hose and hose and coupling assemblies.

3.4.3.1 Hydraulic hose. Hydraulic hose shall be not more than four calendar quarters old at the time the hose and coupling assemblies are delivered to the Government.

3.4.3.2 Hose and coupling assemblies. When tested in accordance with 4.5.2.3 the hose assemblies shall withstand a 15,000 psi proof pressure without permanent deformation, rupture of any part, or external leakage.

3.4.3.3 Hose and coupling assemblies, male and female. Hose and coupling assemblies, male and female, shall withstand 15,000 psi proof pressure without permanent deformation, rupture of any part, or external leakage when tested in accordance with 4.5.2.3.

3.4.4 Cable tensioning assembly. The cable tensioning assembly shall be assembled in accordance with drawing 13217E2193 and when tested in accordance with 4.5.2.4 shall show no evidence of external leakage, (except weepage normally expected at piston rod packing), permanent deformation, or rupture of any part.

3.4.4.1 High and low temperature. The cable tensioning assembly shall withstand high and low temperature testing as specified in 4.5.2.6 without evidence of equipment malfunction, external leakage of a hydraulic component or permanent deformation or damage.

3.4.5 Hydraulic couplings and fittings. Hydraulic couplings and fittings including bushings, flow controls, pipe nipples, elbows, and gauge adapters shall show no evidence of external leakage permanent deformation, or rupture of any part when tested in accordance with 4.5.2.3.

3.4.6 Hydraulic hand pump. The hydraulic hand pump shall be assembled in accordance with drawing 13217E2198 and shall show no evidence of external leakage, permanent deformation, or rupture of any part when tested in accordance with 4.5.2.5.

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3.5 Sealing of threaded hydraulic connections. Each connection of hydraulic fittings with pipe threads shall be sealed with polytetrafluorethylene tape.

3.6 Carrying chests. When all erection equipment shown on TA13217E2171 or all maintenance equipment shown on TA13217E2172 is specified (see 3.1), the following chests shall be furnished:

Erection Equipment, Chest 1 (Two Required)

Drawing	Quantity	Description
13220E8911	1 each	Assembly, Power Unit
13220E6842	1 each	Hose and Coupling Assembly, Female
13217E2205-2	1 each	Hose and Coupling Assembly, Male

Erection Equipment, Chest 2 (Two Required)

Drawing	Quantity	Description
13217E2195	3 each	Cylinder Assembly, Hydraulic, Double Acting
13217E2196-1	2 each	Hose and Coupling Assembly, Female
13217E2205-1	2 each	Hose and Coupling Assembly, Male

Maintenance Equipment, Chest 3 (One Required)

Drawing	Quantity	Description
13217E2195	1 each	Cylinder Assembly, Hydraulic, Double Acting
13217E2190	1 each	Adapter
13220E6842	1 each	Hose and Coupling Assembly, Female
13217E2205-2	1 each	Hose and Coupling Assembly, Male
13217E2198	1 each	Pump Assembly, Hydraulic, Manual
13218E9858	1 each	Nut, Split
13218E9859	1 each	Sleeve

Unless otherwise specified, carrying chests for the following separately procured components shall be furnished:

a. Assembly Power unit (drawing 13220E8911), chest no. 1 as specified above.
b. Cylinder assembly, hydraulic, double acting (drawing 13217E2195), chest no. 2 as specified above, including sufficient space for hose and coupling assemblies.

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Each chest shall separately accommodate the components in as compact a manner as possible. The contractor shall provide adequate blocking and bracing within the chest to insure that the chests will withstand the test specified in 4.6.1.2 without damage to or shifting of any item within the chest. When specified (see 6.2), chests shall not be furnished.

3.7 Instruction plates. Each hydraulic cylinder, hydraulic power unit, and manual hydraulic pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the equipment or components. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material .020- or .032-inch thick. Plates shall be attached by screws or rivets in conspicuous places.

3.8 Stenciling. Each component, except pins, bolts, and washers, shall be stenciled with the following information:

National Stock Number
Nomenclature
Contract Number

Stenciling shall be in accordance with MIL-T-704 and shall be Gothic capitals and numerals, 2 inches high where space permits. Stenciling on small components shall be as large as practicable.

3.9 Metal stamping. In addition to stenciling, the national stock number, nomenclature and contract number shall be metal stamped in an exposed location near one end of each component. The letters and numerals shall be 1 inch high where space permits. On small components, the metal stamping shall be as large as practicable.

3.10 Identification marking for hose assemblies. Each hose assembly shall have identification marking metal die stamped on an aluminum band conforming to MIL-B-43436, except that the band shall be of sufficient width to allow a minimum of three lines of 1/8-inch metal stamping. The marking band shall contain the following information:

National Stock Number
Nomenclature
Mfg Code _____ Date of Mfg _____

3.11 Erection equipment. The cable reinforcement erection equipment shall conform to top assembly TA13217E2171. When tested as specified in 4.5.2.6 and 4.5.2.7, the erection equipment shall show no evidence of external leakage of a hydraulic component or evidence of permanent deformation or damage.

3.12 Maintenance equipment. The cable reinforcement maintenance equipment shall conform to TA13217E2172. When tested as specified in 4.5.2.6, the maintenance equipment shall show no evidence of external leakage of a hydraulic component and shall show no evidence of permanent deformation or damage.

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3.13 Treatment and painting. The erection and maintenance equipment shall be treated and painted in accordance with MIL-T-704, type F. Pins, bolts, nuts, and threaded surfaces shall not be painted but shall be coated with a preservative conforming to MIL-C-16173, grade 2.

3.14 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.2):

Item No.	Nomenclature	National Stock Number	Quantity for each
1.	Bridge, Fixed, Steel Panel, Bailey Type M-2.	5420-00-530-3784	1
2.	Bridge Erection Set, Fixed Bridge, Bailey Type, M-2.	5420-00-530-3785	1
3.	Truck, Dump, 5 Ton, 6 X 6, M-51 (or equivalent).	2320-00-835-8336	30
4.	Trailer, basic utility, pole type, 2-1/2 T, 2 whl (or equivalent).	2330-00-697-8102	9

3.15 Workmanship.

3.15.1 Metal fabricating. The surfaces of all complete members used in fabrication shall be free from buckling, kinks, wrinkles, washboarding, burrs, scribe lines, and scratches. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Moving parts shall operate without binding. The straightening of material shall be done by methods that will not cause injury to the material.

3.15.2 Welders or welding operators. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who make only horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.8).

3.15.3 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the erection and maintenance equipment to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought

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into as close contact as possible and in no event shall be separated by more than 1/16 of an inch, unless appropriate bridging techniques are used. Welding procedures for structural steel shall be qualified in accordance with applicable ASME or AWS code, and welding and welds of structural steel shall be in accordance with ASME or AWS code.

3.15.3.1 High strength low-alloy steel and heat treated high strength steel. Welding of high strength steel and heat treated high strength steel shall be in accordance with MIL-STD-1261, class 3. The workmanship specimens required by MIL-STD-1261 shall be samples simulating all welding conditions.

3.15.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.15.5 Machine work. Tolerances and gauges for metal fits shall conform to the limitations specified on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP)

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shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The erection equipment, maintenance equipment, or separately procured components as applicable shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The erection equipment, maintenance equipment, or separately procured components as applicable shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.3.2.1 Hydraulic-system components. Prior to installation of the hydraulic-system components into the first article, the components shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, the erection equipment shown on TA13217E2171, the maintenance equipment shown on TA13217E2172, or individual components of either of the data lists as specified (see 3.1) shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination and tests of erection equipment, maintenance equipment, and separately procured components shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found, and rejected when one or more defects are found.

4.4.3 Examination.

4.4.3.1 Individual. Each component of the erection equipment, maintenance equipment and each separately procured component shall be examined for defect 101 as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.2 shall be examined for defects 102 through 202 as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.4 Tests.

4.4.4.1 Individual. Each hydraulic cylinder shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3(d). Failure of any test shall be cause for rejection.

4.4.4.2 Samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3(a), (b), (c), and (e), 4.5.2.4, and 4.5.2.5. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The erection equipment, maintenance equipment, or separately procured components shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the QAP requirements as shown on the drawings (see 3.1).
102. Components missing or not as specified (see 3.1).
103. Assembly incomplete or incorrect (see 3.1).
104. Any drawing requirement, other than those identified by QAP data, not as specified (see 3.1.1).
105. Materials not as specified (see 3.3).
106. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.3.1).
107. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
108. Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.3.1.2).
109. Age of hydraulic hose not as specified (see 3.4.3.1).
110. Workmanship not as specified (see 3.15).
111. Qualification of welders or welding operators not as specified (see 3.15.2).
112. Welding not as specified (see 3.15.3).
113. Bolted connections not as specified (see 3.15.4).

Minor

201. Identification and other required markings not as specified (see 3.7, 3.8, 3.9, and 3.10).
202. Cleaning, treatment, and painting not as specified (see 3.13).

4.5.2 Tests.

4.5.2.1 Test conditions. All tests, except 4.5.2.6, shall be conducted at an ambient temperature between +40 °F and 90 °F. Hydraulic oil temperature shall not exceed +190 °F at the pump discharge during testing. The equipment shall operate as specified herein without maintenance.

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4.5.2.2 Cleanliness of hydraulic components. Prior to assembling the hydraulic system, clean and flush each hydraulic component. Prior to flushing, position each cylinder in such a manner that entrapment of contaminants is held to a minimum. Presence of particles on a 100-micron filter screen shall constitute failure of this test.

4.5.2.3 Proof pressure. Subject the following components to a proof pressure of 15,000 psi ±100 psi for 5 minutes:

a. 13217E2206 - 1 and 2 - Hose Assembly.
b. 13217E2196 - 1 and 2 - Hose and Coupling Assembly, Female.
c. 13217E2205 - 1 and 2 - Hose and Coupling Assembly, Male.
d. 13217E2195 - Cylinder and Assembly Hydraulic Double Acting.
e. Individual components detailed on the drawings, such as, but not limited to, nipples, flow control valves, bushings, and quick-disconnect couplings.

The proof pressure shall be applied alternately to each port of the double acting hydraulic cylinder assembly (drawing 13217E2195).

4.5.2.4 Cable tensioning assembly. Assemble the cable tensioning assembly as specified in 3.4.4 and cycle the assembly a minimum of 25 complete cycles. After completion of 25 cycles, subject the assembly to a pressure of 9500 psi ±100 psi and hold the pressure for 20 minutes. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.5 Hydraulic hand pump. The hydraulic hand pump shall be tested for conformance with performance criteria shown on the drawings. Failure to conform to the drawing requirements shall constitute failure of this test.

4.5.2.6 Hydraulic system environmental test. The hydraulic equipment, as follows, shall be subjected to the tests contained in 4.5.2.4 under temperature conditions of +125 °F ±5 °F and -25 °F ±5 °F:

a. Cable tensioning assembly.
b. Maintenance equipment.

4.5.2.6.1 Failure criteria. Failure of equipment to function at the temperature limits specified or evidence of external leakage of a hydraulic component or permanent deformation or damage shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. Examine the first article pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

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4.6.1.2 Test. The first article pack of crated erection equipment, maintenance equipment, or separately procured components for level A shall be subjected to the railcar impact test in accordance with MIL-STD-1186, appendix A. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the equipment or the chests and boxes shall constitute failure of the first article pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects.

114.	Unfinished exterior metal surfaces not coated with preservative as specified for level A (see 5.2.1.2.1).
115.	Machined surfaces not coated with preservative and wrapped and secured as specified for level A (see 5.2.1.2.2).
116.	Cable tensioning assembly not preserved as specified for level A (see 5.2.1.3).
117.	Hydraulic maintenance equipment not preserved as specified for level A (see 5.2.1.4).
118.	Consolidation not as specified for level A (see 5.2.1.5).
119.	Blocking, bracing, and anchoring not in accordance with the referenced documents for level A (see 5.2.1.5).
120.	Packing not as specified for level A and C (see 5.3).
121.	Marking missing, illegible, incorrect, or incomplete for level A or C (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model or production item. If the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model and the production items.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Unprotected surfaces.

5.2.1.2.1 Unfinished surfaces. Unpainted or unfinished exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.2.2 Machined surfaces. Exposed machined metal surfaces shall be coated with type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, grade A, type I or II, class 2, and material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.3 Cable tensioning assembly.

5.2.1.3.1 Hydraulic cylinders. Each hydraulic cylinder shall be filled with hydraulic oil (see 3.3.3) and the piston fully retracted into the cylinder and the cylinders placed in a chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.3.2 Hose and coupling assembly. Each hose and coupling assembly shall be coiled and secured with cotton tape or twine and placed in a chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.3.3 Hydraulic power unit. Each hydraulic power unit supply reservoir shall be filled with hydraulic oil (see 3.3.3) and placed in the chest or box as specified (see 3.6 and 5.2.1.5).

5.2.1.4 Hydraulic maintenance equipment.

5.2.1.4.1 Manual pump assembly. Each manual pump assembly shall be preserved and placed as specified in 5.2.1.3.1.

5.2.1.4.2 Hose and coupling assembly. Each hose and coupling assembly shall be coiled, secured, and placed as specified in 5.2.1.3.2.

5.2.1.5 Consolidation. When chests are provided (see 3.6) the components shall be placed and secured in the appropriate chest and the chest lid secured. When chests are not provided, the components shall be placed and secured in close-fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The contents shall be blocked, braced, anchored, or cushioned as required to prevent movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required. When the individual components are procured as replacements or spares, they shall not require consolidation but shall be directly packed as specified in 5.3.1.

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5.2.2 Level C. The equipment shall be preserved as specified for level A except the boxes shall be domestic type or class, whichever is applicable.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. The equipment contained in the appropriate chests or boxes shall be packed together in a crate conforming to MIL-C-52950, type V, style A, nondemountable. The chests or boxes shall be secured to the crate base in a manner to prevent movement or damage in accordance with the appendix to the crate specification. When individual components of the erection and maintenance equipment are procured as replacements or spares, the components shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated size as applicable and ASTM D 4675.

5.3.2 Level C. The equipment preserved as specified in 5.2, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rule or National Motor Freight Classification Rules.

5.4 Marking. In addition to any special marking specified in contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The erection and maintenance equipment is intended to be used to attach the cable reinforcement equipment covered by MIL-E-52714(ME) to the triple single configuration of the M-2 panel bridge (Bailey type). The equipment also provides the means of checking and adjusting the post tension in the bridge. This item is acomponent of the following supply catalog: SC5420-97-CL-E50.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Whether the contractor shall furnish all the erection equipment on TA13217E2171, all the maintenance equipment on TA13217E2172, or separately procured components selected from either of the two data lists (see 3.1).
d. Time frame required for submission of the first article and number of units of equipment or separately procured components required (see 3.2).

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e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some or all of the first article examination and tests, the contracting officer should specify which examination and tests shall be conducted by the contractor (see 3.2).
f. When first article inspection is required and the number of units of equipment or separately procured component, to be furnished (see 3.2).
g. When carrying chests are not required (see 3.6).
h. Time frame for submission of the first article pack (see 5.1).
i. Level of preservation and level of packing required (see 5.2 and 5.3).
j. Any special marking (see 5.4).

6.3 First article. When a first article inspection is required, the item(s) should be an initial production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First article pack. Any changes or deviations of production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the equipment and components in accordance with this specification.

6.5 Quality assurance provisions (QAP). The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrected action taken when applicable.

6.5.1 Definition. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance manual to be furnished with the cable reinforcement equipment.

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6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions regarding shipment of the equipment covered by this specification.

6.8 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.15.2	DI-MISC-80876	Welding Procedure Qualification Test Report
3.15.2	DI-MISC-80875	Welding Procedures

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject term (key word) listing.

Bridge, panel (Bailey type)
Cable reinforcement set
Erection and maintenance equipment

Custodian:
Army - ME

Review activity:
DLA-CS

Preparing activity:
Army - ME

Project 5420-0186

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-52715C(ME)	2. DOCUMENT DATE (YYMMDD) 92/05/04

3. DOCUMENT TITLE
Erection and Maintenance Equipment for Cable Reinforcement Set For Panel Bridge, M-2 (Bailey Type)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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